

Concept Note on the study: Global Computational Re-analysis of Gut-microbiome in pregnant women with Tuberculosis.

1. Background / Context

Tuberculosis (TB) remains one of the leading causes of morbidity and mortality among women of reproductive age globally, with pregnancy-associated Tuberculosis posing additional risks to both maternal and foetal health (reference). TB during pregnancy is associated with increased risks of maternal *anaemia*, *preterm birth*, *low birth weight*, and *adverse neonatal outcomes*, particularly in high-burden, resource limited settings (references). Despite these risks, pregnant women are frequently under-represented or analytically combined with non-pregnant adults in TB research, limiting biological specificity (reference).

Pregnancy represents a distinct physiological state characterised by tightly regulated immunological, metabolic, and hormonal adaptations (reference). These include a shift towards immune tolerance, and metabolic remodelling to support foetal development. Such changes fundamentally differentiate pregnant individuals from non-pregnant adults and are likely to influence host–microbiome interactions (reference).

The gut microbiome plays a critical role in immune regulation, metabolic homeostasis, and drug metabolism (reference). In TB, both infection and Anti-Tubercular Therapy (ATT) have been shown to perturb gut microbial communities, with downstream effects on immune function and systemic inflammation (references). Understanding gut microbiome dynamics during TB in pregnancy is therefore biologically and clinically relevant, particularly given the combined effects of infection, antibiotic exposure, and pregnancy-associated immune modulation.

2. Problem Statement / Knowledge Gap

Duvallet et al. (2017) demonstrates that many disease-associated microbiome changes are non-specific stress responses, Established computational re-analysis as the gold standard, and shows that collapsing heterogeneous studies can reveal shared dysbiosis signatures.

Yang et al. (2020) Establishes pregnancy as a stable, regulated microbiome state, and also stresses that gestational age explains little microbiome variance, and lastly Identifies host heterogeneity as the dominant driver in healthy pregnancy.

Baral et al. (2025) Integrates TB, gut microbiome, and host immunity computationally, Moving beyond association to immune–amicrobiome interaction modelling, and established that TB is a systemic disease affecting the gut. This segment should generally not be limited to only three references. Other similar works should be stated.

All three papers stop at the same biological frontier, though from different angles:

- Duvallet tells us what disease dysbiosis looks like globally
- Yang tells us what pregnancy microbiomes look like when healthy
- Baral tells us how TB interacts with the microbiome via immunity

Gap

Briefly state the current information related to TB-lung-gut microbiome and TB-vaginal microbiome available (some examples in these references- <https://www.nature.com/articles/nrmicro.2016.142>, <https://nyaspubs.onlinelibrary.wiley.com/doi/10.1111/nyas.14300>, <https://linkinghub.elsevier.com/retrieve/pii/S2352340918312058>)

However, no study has computationally examined how tuberculosis-associated gut microbiome alterations manifest within the distinct immunological and physiological context of pregnancy.

As a result, it remains unclear whether pregnancy preserves, attenuates, or reshapes the specific dysbiosis patterns observed in tuberculosis, or whether pregnancy introduces unique microbiome-immune interactions that are not captured in existing adult-focused analyses.

There is a need to highlight what is scientifically missing, why it matters, and what exactly is unknown for pregnant women with TB.

Public microbiome repositories contain TB- and pregnancy-related datasets, yet no dedicated computational studies have leveraged these resources to characterize the gut microbiome of pregnant women with tuberculosis.- this is the gap this to fill.

This absence of pregnancy-specific computational re-analysis represents a critical gap in the interpretation of tuberculosis-microbiome interactions.

3. Rationale for the Study

Publicly available microbiome datasets (provide some examples here) represent an underutilised resource for addressing this gap. Advances in computational biology now allow for harmonised reprocessing of heterogeneous sequencing data, adjustment for study-level effects, and robust confounder control (references).

A computational re-analysis focused specifically on pregnant women with TB is both feasible and scientifically justified. It enables pregnancy-aware interpretation of existing TB microbiome data without the logistical, ethical, and financial constraints of primary data collection. This approach builds directly on established meta-analytic and integrative frameworks while addressing a clearly defined biological blind spot.

4. Aim of the Project

To determine whether the immune-tolerant physiological state of pregnancy modifies tuberculosis-associated gut microbiome dysbiosis patterns through a harmonised computational re-analysis of publicly available global microbiome datasets.

1. To identify and curate publicly available gut microbiome datasets containing pregnant individuals with tuberculosis or anti-tubercular therapy exposure, and to assess metadata completeness and study-level heterogeneity.
2. To evaluate the influence of key biological and clinical covariates, including gestational age, HIV status, antibiotic exposure, and study-level effects, on observed microbiome alterations through multivariable and sensitivity analyses.
3. To compare tuberculosis-associated gut microbiome alterations observed in pregnancy with established adult TB dysbiosis patterns in order to assess conformity to, or divergence from, non-specific disease-associated signatures.
4. To quantify gut microbiome diversity and taxonomic shifts associated with tuberculosis and anti-tubercular therapy during pregnancy using harmonised computational pipelines.

6. Study Design and Data Sources

This project will employ a secondary data analysis design. Only publicly available 16S rRNA gene sequencing or shotgun metagenomic datasets will be included. Data will be sourced from established repositories.

As all data are publicly available and de-identified, no additional ethical approval is required for participant recruitment.

7. Metadata and Inclusion Criteria

Mandatory metadata fields include:

- Pregnancy status
- Tuberculosis status and/or anti-tubercular therapy exposure

Additional variables to be extracted, where available, include:

- Gestational age or trimester
- HIV status
- Antibiotic exposure (including ATT phase)
- Geographic region or study site

You may include non-pregnant individuals in the inclusion criteria as a comparator group

A formal metadata completeness assessment will be conducted prior to inclusion to ensure analytical validity.

8. Analytical Approach

This study will employ a harmonised, confounder-aware computational re-analysis framework designed for cross-study microbiome integration. All analyses be conducted using reproducible workflows implemented in QIIME 2 and R, incorporating established methods such as PERMANOVA, ANCOM-BC, and MaAsLin2, with sensitivity analyses using alternative differential abundance and batch-adjustment approaches.

phase 1: Dataset Identification, Curation, and Study-Level Assessment

Publicly available gut microbiome datasets will be systematically identified from international repositories, including the NCBI Sequence Read Archive. Inclusion will be restricted to datasets with explicit pregnancy status and tuberculosis infection and/or anti-tubercular therapy exposure.

A structured metadata audit will be conducted before analysis to assess the availability and completeness of key covariates, including gestational age, HIV status, antibiotic exposure, sequencing platform, and geographic region. Study-level characteristics will be summarised to inform downstream modelling and to guide the inclusion of random or fixed effects in statistical analyses.

Phase 2: Pregnancy-Specific TB Microbiome Characterisation

Raw sequencing data will be reprocessed using standardised pipelines to generate comparable taxonomic profiles across studies.

Within-sample diversity will be assessed using established alpha diversity indices, while between-sample community differences will be evaluated using beta diversity distance metrics. Community-level differences associated with TB and anti-tubercular therapy during pregnancy will be tested using **PERMANOVA**, with study identifier included as a stratification or blocking factor to account for inter-study heterogeneity.

Differential taxonomic abundance will be assessed using **ANCOM-BC** and **MaAsLin2**, both of which are widely used compositionality-aware methods suitable for multi-cohort microbiome analyses.

Phase 3: Comparative Assessment Against Adult TB Dysbiosis Patterns

Pregnancy-associated TB microbiome signatures will be evaluated in the context of established adult TB dysbiosis patterns reported in prior large-scale re-analyses. Comparative analyses will focus on directionality and consistency of taxonomic shifts rather than exact effect sizes.

phase 4 : Confounder Adjustment and Sensitivity Analyses

Multivariable models will be used throughout to adjust for key biological and clinical covariates, including gestational age, HIV status, antibiotic exposure, and geographic region where available. Study identifier will be incorporated as a covariate or random effect to control for study-specific biases.

Software Ecosystem

Primary Analysis

- phyloseq – for microbiome data handling
- vegan – for ecological statistics (PERMANOVA, diversity)
- MaAsLin2, ANCOMBC, ALDEx2 – for differential abundance
- lme4 – for mixed-effects modelling

9. Expected Outcome

1. Curated and Harmonised Dataset

- A systematically curated collection of publicly available gut microbiome datasets including pregnant individuals with tuberculosis or anti-tubercular therapy exposure.
- Comprehensive metadata assessments highlighting completeness, study-level heterogeneity, and potential covariates for multivariable adjustment.

2. Quantification of Pregnancy-Specific TB-Associated Microbiome Changes

- Identification of taxonomic shifts in the gut microbiome associated with tuberculosis and Anti-tubercular therapy during pregnancy.
- Statistically robust community level differences, accounting for study specific effects and confounders, providing insight into microbiome resilience or perturbation during gestation.

3. Comparative Insights Against Established Adult TB Dysbiosis Patterns

- Determination of whether pregnancy modifies typical adult TB-associated gut dysbiosis signatures.
- Insights into the potential influence of immune tolerance during pregnancy on TB related microbiome alterations.

4. Methodological and Computational Framework Outputs

- Reproducible pipelines integrating QIIME 2, phyloseq, vegan, MaAsLin2, ANCOM-BC, ALDEx2, and lme4 for cross-study microbiome re-analysis.
- A transparent, reproducible analytical framework enabling future integration of new datasets and validation of pregnancy-associated microbiome hypotheses.

5. Scientific Dissemination

- A manuscript draft suitable for submission to a peer-reviewed journal, presenting key findings, methodological rigour, and biological insights.
- Contribution to global microbiome research through open-science principles, supporting reproducibility.

10. Significance and Contribution

This study will provide the first pregnancy-focused computational synthesis of gut microbiome alterations associated with tuberculosis. The findings will inform the design of future experimental and clinical studies, support pregnancy modelling of TB–microbiome dynamics, and fill a critical gap in global TB microbiome research.

In summary, this study will not only deepen mechanistic insights into pregnancy modified tuberculosis microbiome interactions but will also contribute a reproducible methodological framework, curated multi-cohort datasets, and actionable evidence for clinical and research applications in maternal health and infectious diseases.